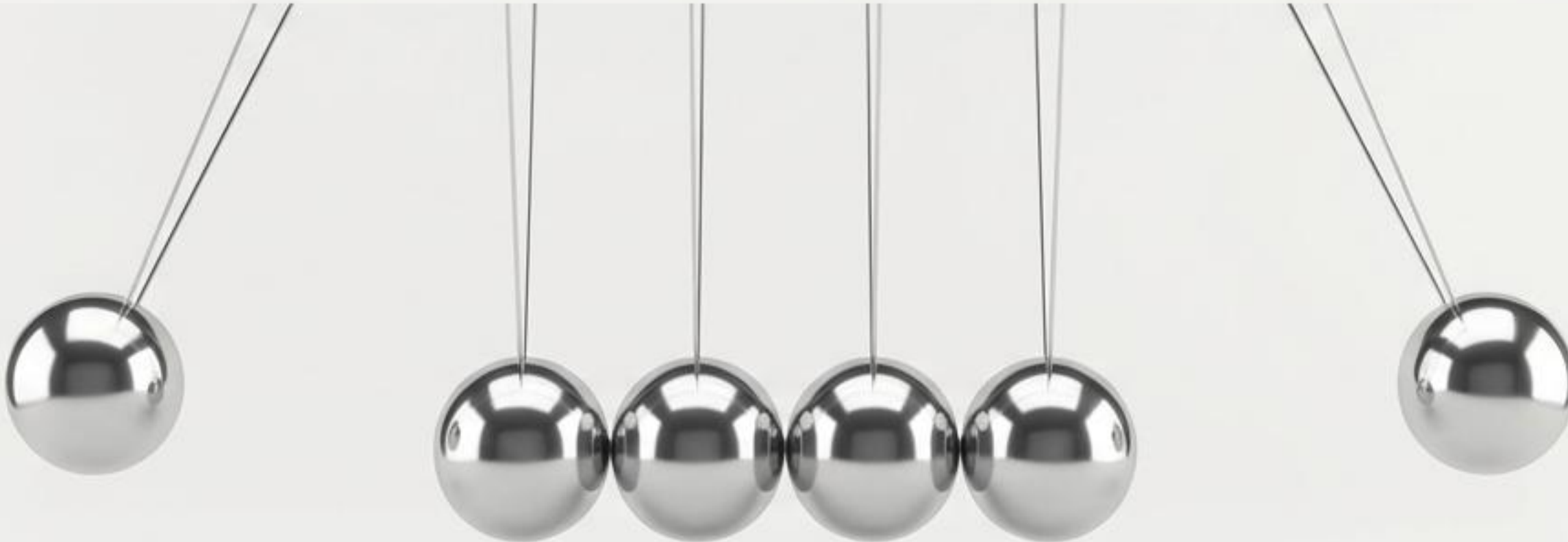


The Quant's Compass: A Strategic Guide to Mean Reversion

From Statistical Theory to Practical Implementation

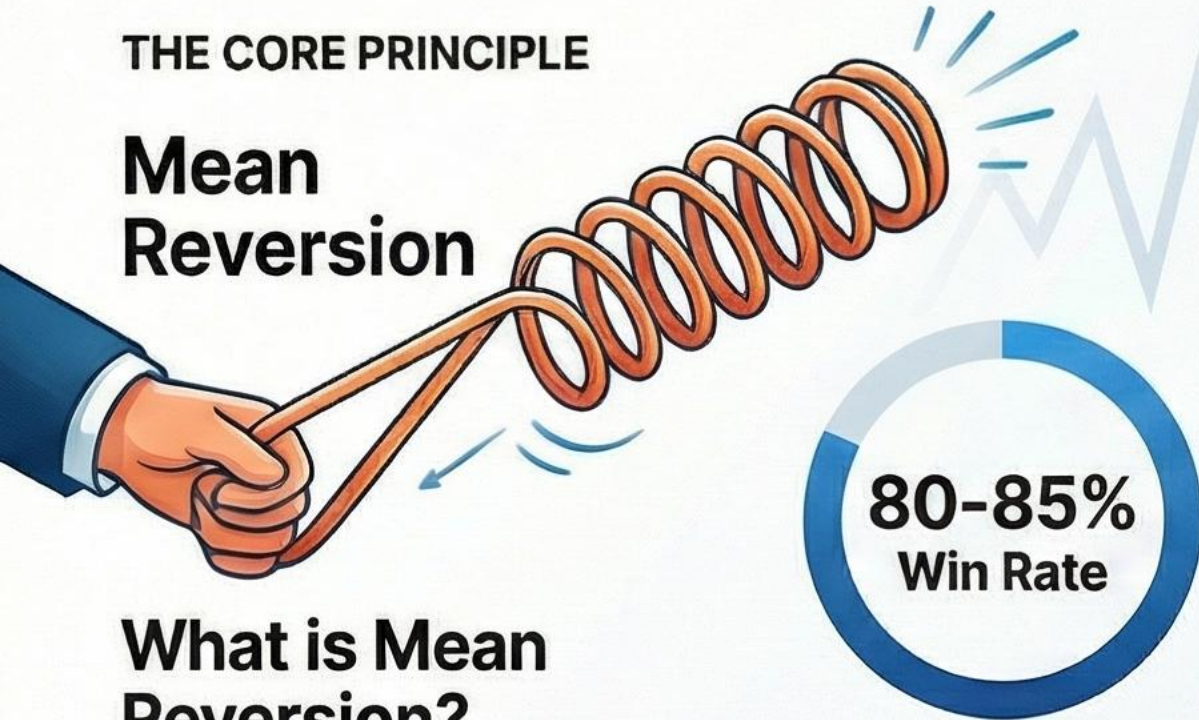
By Michael Fernandes



Trading the Snap-Back: An Introduction to Mean Reversion

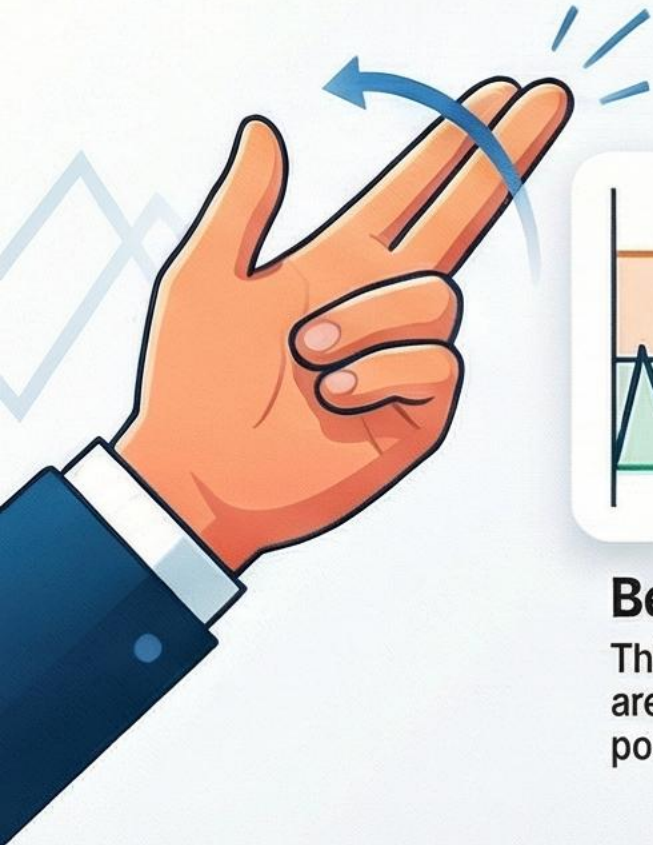
THE CORE PRINCIPLE

Mean Reversion



What is Mean Reversion?

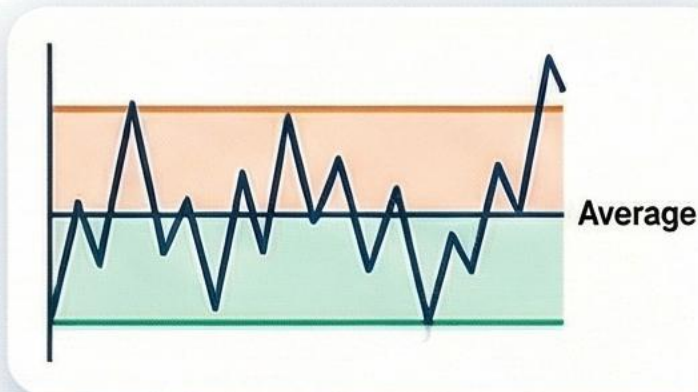
The theory that prices revert to their historical average after a significant deviation.



80-85% Win Rate

80-85% Win Rate

Strategies typically result in a high percentage of profitable trades, but with small average gains.



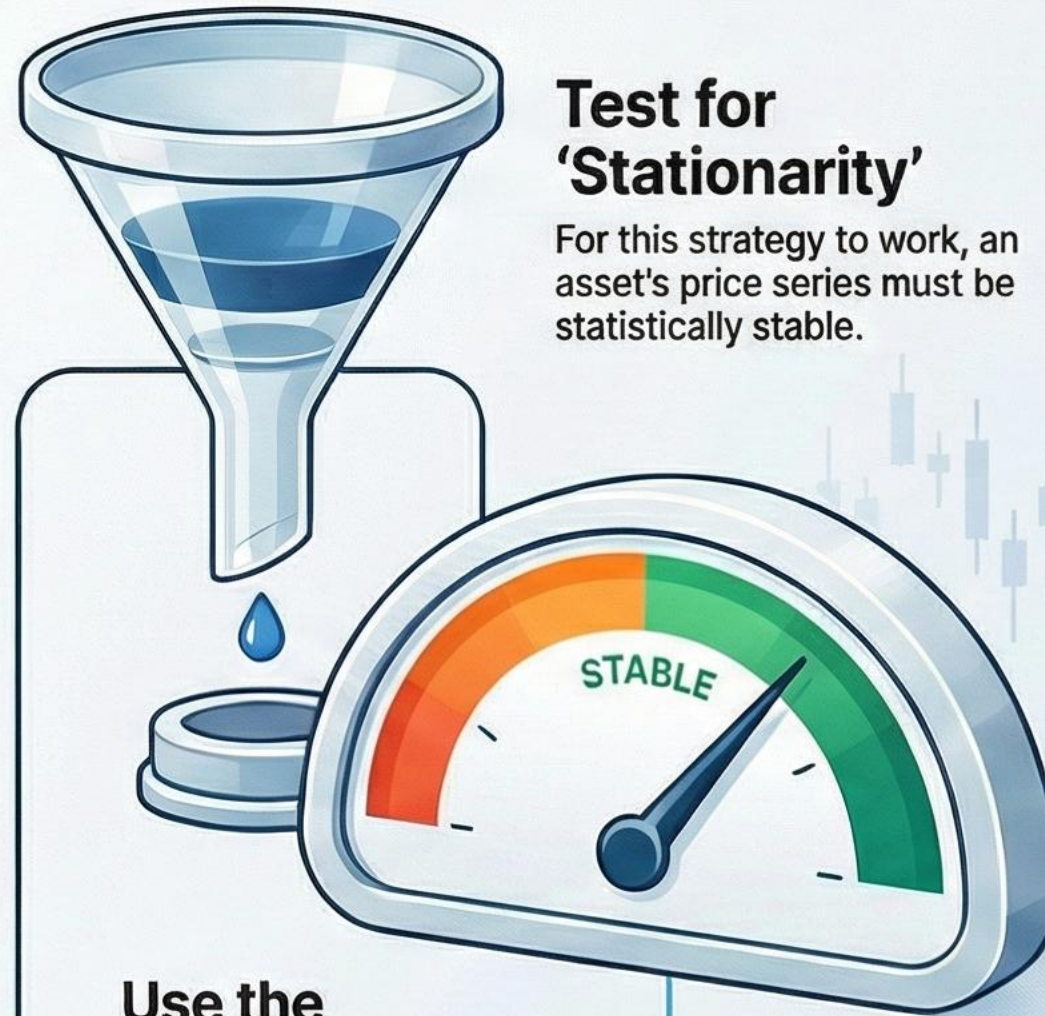
Best in Sideways Markets

This approach excels when markets are range-bound and performs poorly during strong trends.

HOW TO FIND MEAN-REVERTING ASSETS

Test for 'Stationarity'

For this strategy to work, an asset's price series must be statistically stable.



Use the Augmented Dickey-Fuller (ADF) Test

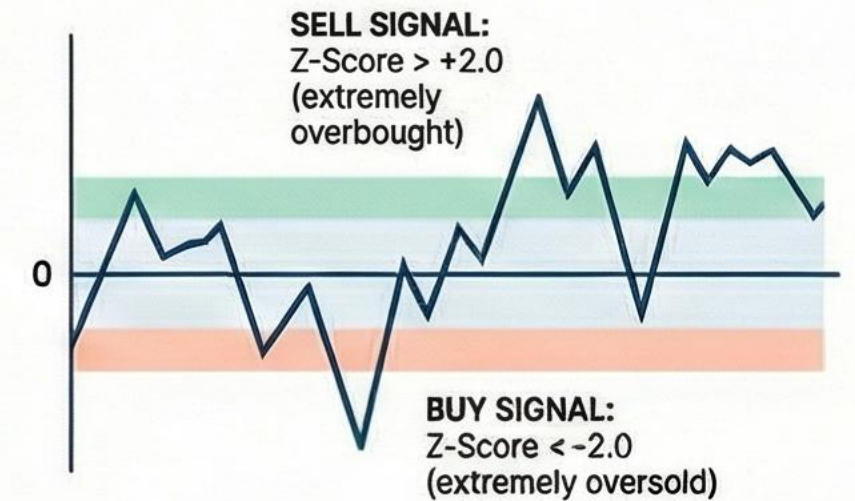
This is the standard statistical test to confirm if a time series is mean-reverting.



Look for a p-value < 0.05

This result gives 95% confidence that the asset is a good candidate.

HOW TO TRADE IT: COMMON STRATEGIES



The Z-Score Method

This measures how many standard deviations a price is from its rolling average.



Bollinger Bands

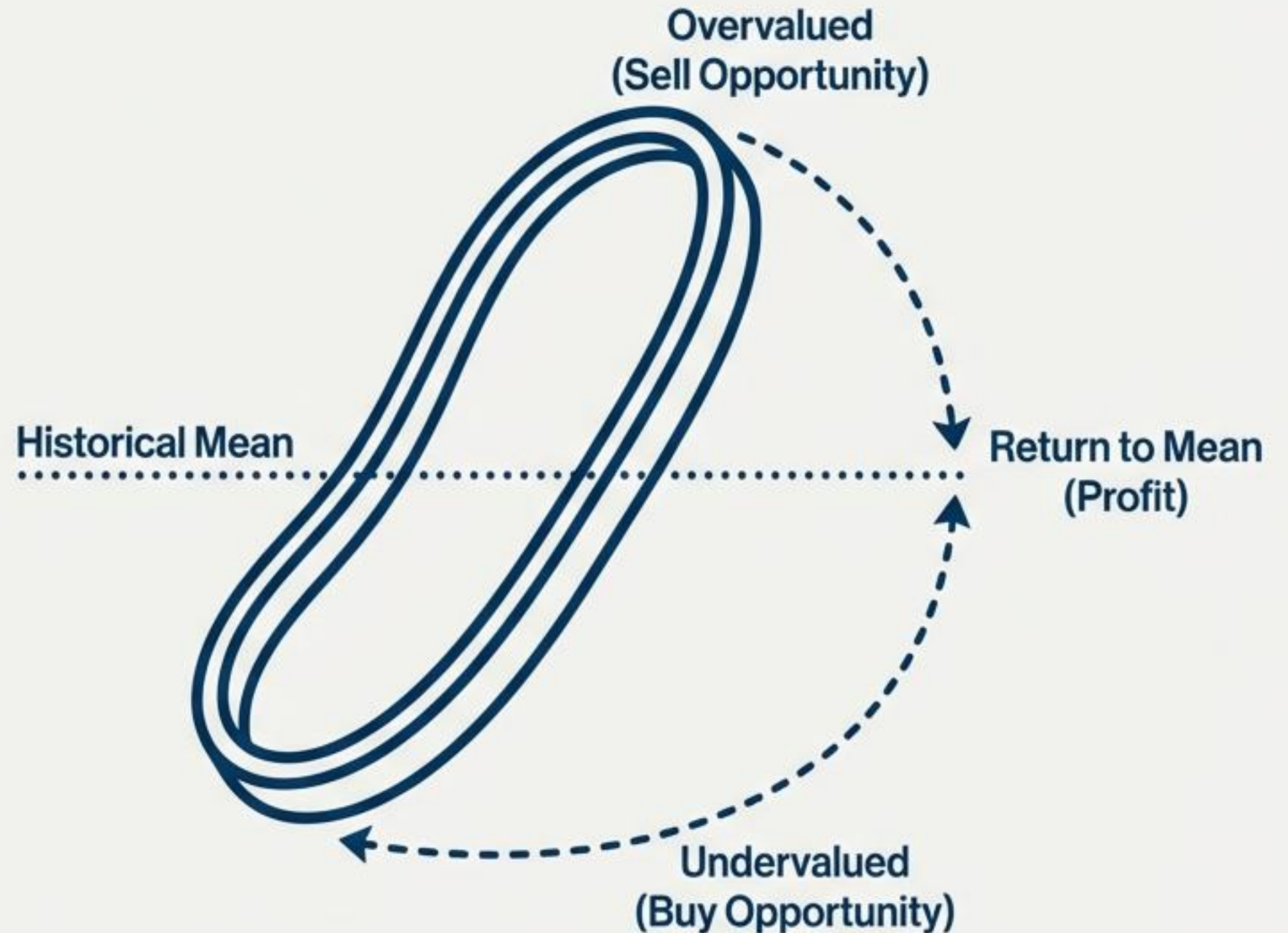
This creates a dynamic price channel that adapts to market volatility.

The Core Principle: Prices Revert to an Equilibrium

Core Principle: Asset prices and returns eventually revert to their long-term historical average after significant deviations.

Trading Logic: Buy undervalued assets trading below their mean; Sell overvalued assets trading above their mean. Profit is realised when prices return to equilibrium.

Why it Works: Extreme prices trigger corrective mechanisms. Low prices attract buying demand, while high prices trigger selling pressure, creating oscillations around a fundamental value.



A Performance Profile Built on High-Probability, Modest Wins



80-85%

Win Rate

The majority of trades are expected to end profitably.



High

Trade Frequency

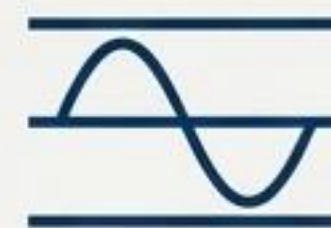
Numerous opportunities arise from identifying price deviations.



Small & Frequent

Average Gain

The strategy accumulates profit through many small wins, not occasional large ones.



Range-Bound Markets

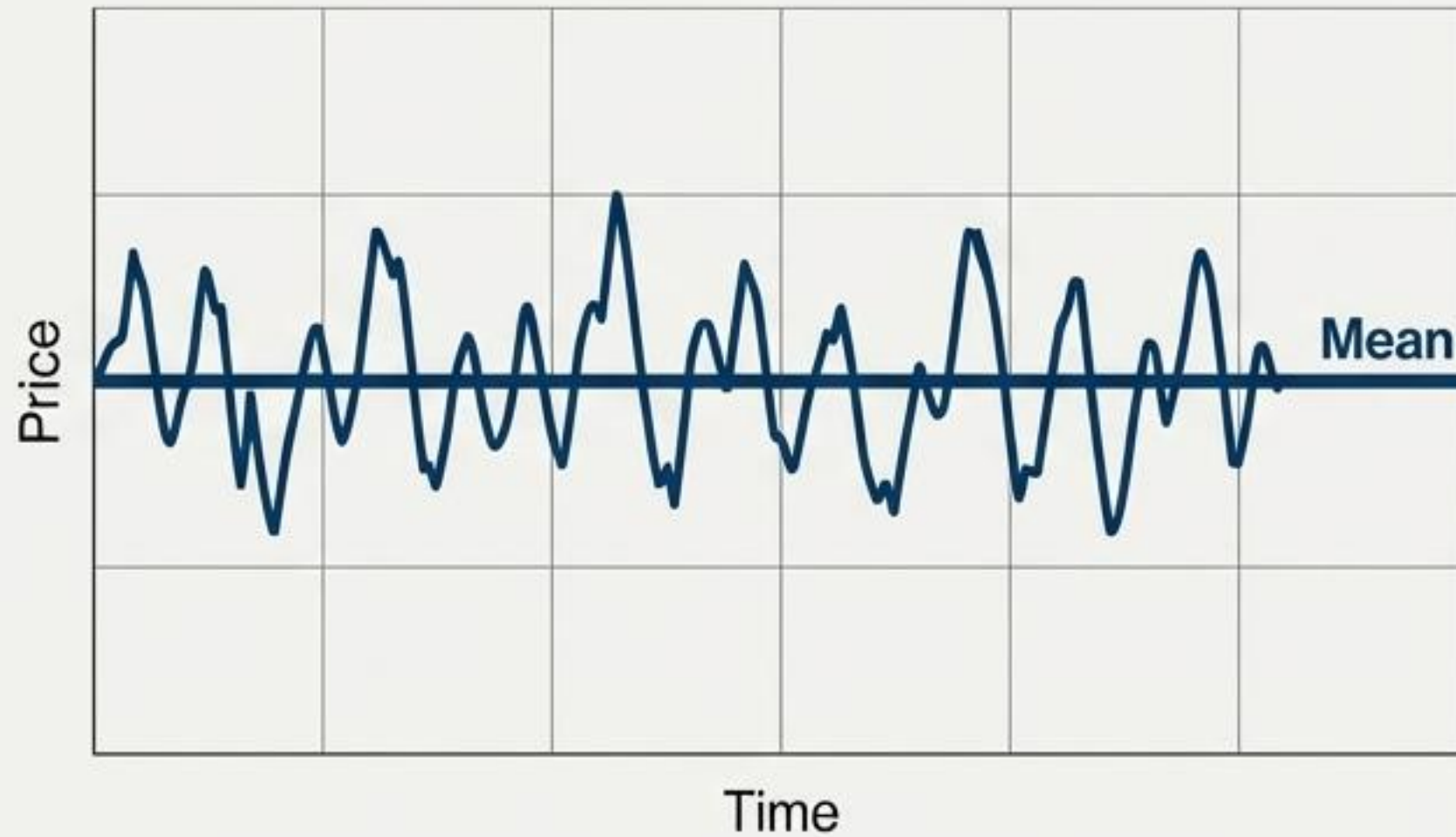
Optimal Environment

Performs poorly in strong, persistent trends.

The Statistical Prerequisite: Mean Reversion Only Works on Predictable Paths

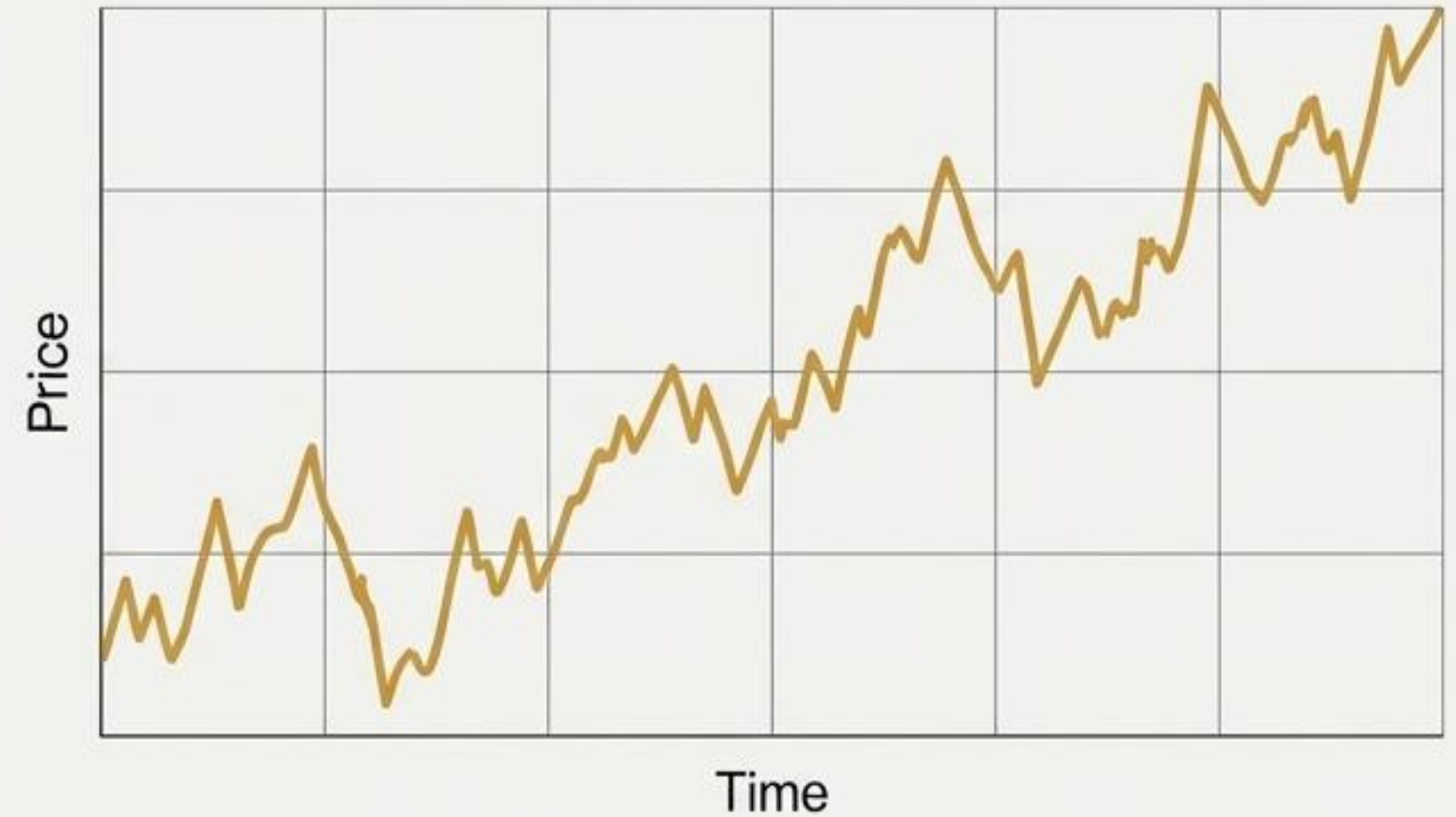
Stationary Series (Valid for Trading): Exhibits constant mean, constant variance, and no predictable drift. Its statistical properties are stable over time.

Mean-Reverting Behaviour



Non-Stationary ‘Random Walk’ (Invalid for Trading): The mean changes over time and variance can grow unbounded. It follows an unpredictable path.

Random Walk



Trading mean reversion on a non-stationary asset is like expecting a ball thrown upward to “revert” downward—it won’t. A random walk continues drifting in unpredictable directions.

Augmented Dickey-Fuller (ADF) Test

Purpose:

To statistically confirm if a time series exhibits mean-reverting behaviour.

Hypothesis Framework:

 **Null Hypothesis (H_0):** The series has a unit root (it is non-stationary).

 **Alternative Hypothesis (H_1):** The series is stationary (it is mean-reverting).

Practical Example:

For a stock price series with an ADF Statistic of -3.45 and a 5% Critical Value of -2.89, we reject the null hypothesis because the statistic is more negative than the critical value.



$p < 0.01$

99% confidence. Strong candidate.
Many funds require this level.

$0.01 < p < 0.05$

95% confidence. Good candidate.
Essential minimum threshold.

$p > 0.05$

Insufficient evidence. Do not trade
as a mean reversion strategy.

The Mathematical Engine: Modelling the 'Pull' Towards the Mean

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t$$

Gravitational Pull

The drift term that pulls the process back towards the long-term mean (μ). The parameter θ represents the speed of this reversion.

Random Shocks

The volatility term that creates temporary, random deviations from the mean.

Practical Application (The Half-Life)



Formula: Half-Life = $\ln(2) / \theta$



Interpretation: This calculates the time required for a price deviation to decay by 50%. A tangible measure of the reversion speed.

Quantifying Deviation from the Mean with Z-Scores and Bollinger Bands

Z-Score Explained

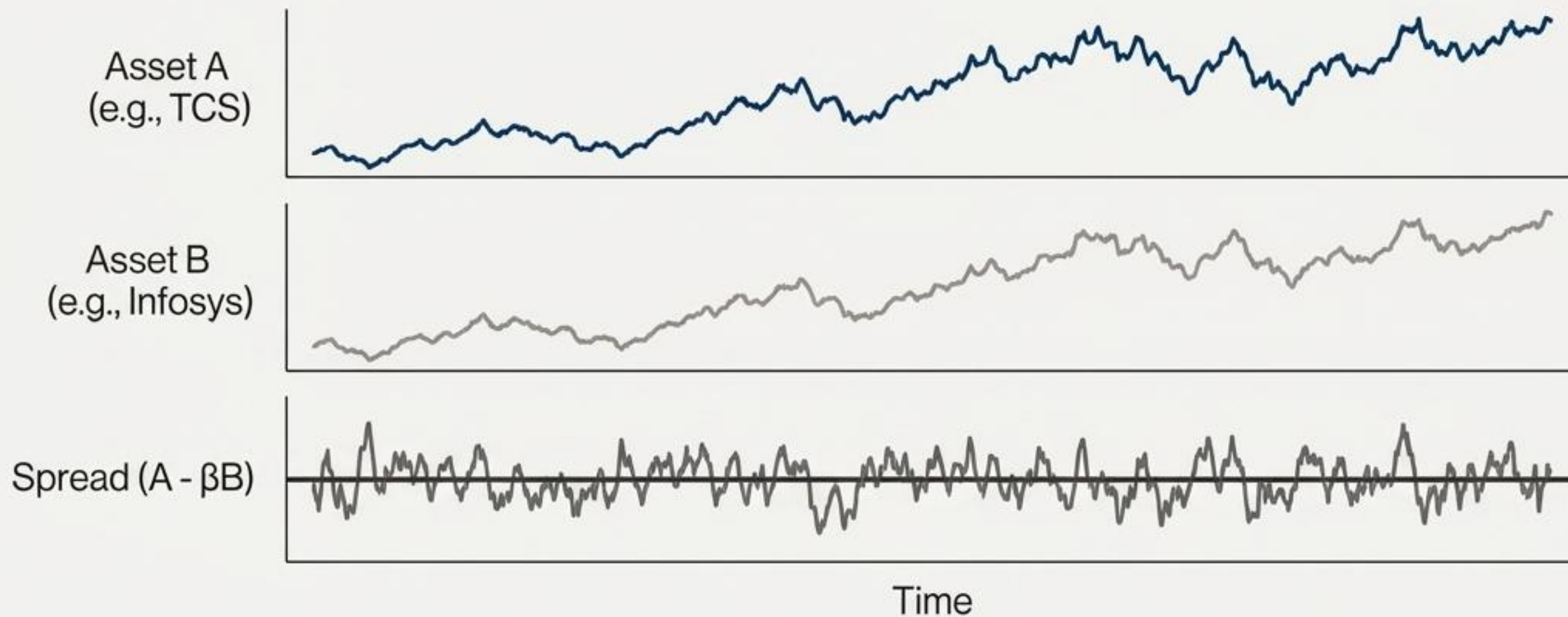
- **Concept:** Measures how many standard deviations a price has deviated from its rolling mean.
- **Formula:**
$$Z = \frac{(\text{Current Price} - \text{Rolling Mean})}{\text{Rolling Standard Deviation}}$$
- **Trading Signals:** 
 - ↑ **Sell/Short Signal:** $Z > +2.0$ (Overbought)
 - ↓ **Buy/Long Signal:** $Z < -2.0$ (Oversold)
 - **Exit Signal:** Position is closed when Z reverts towards 0.



Pairs Trading: A Market-Neutral Approach to Mean Reversion

Core Concept: Instead of trading a single asset, Pairs Trading focuses on the *spread* between two highly correlated assets. The goal is to profit from the temporary divergence and subsequent convergence of their prices.

Key Benefit: It is a market-neutral strategy. By being long one asset and short the other, the position is insulated from broad market movements, profiting only from the relative mispricing between the pair.



The Disciplined Workflow for Pairs Trading



1. Selection

Identify two assets with a strong historical relationship.

Key metric: Correlation > 0.8.

2. Cointegration

Statistically verify that the spread is stationary.

Key tool: Apply the ADF test to residuals.

Requirement: p-value < 0.05.

3. Hedging

Calculate the hedge ratio (β) to ensure the combined position is market-neutral.

4. Trading

Calculate the Z-Score of the spread.

Generate signals when Z-Score crosses predefined thresholds (e.g., +/- 1.5).

Adapting to Shifting Markets with the Kalman Filter

The Problem with Static Models: Traditional tools like fixed hedge ratios assume stable relationships. They fail when market regimes shift, as correlations and volatility are not constant.

The Solution: Kalman Filter: A dynamic model that continuously updates its estimates (like the hedge ratio or mean level) based on new price data, adapting automatically to changing market conditions.

Applications: Dynamically updating hedge ratios, tracking a moving mean of a spread, and estimating true volatility.



Choosing the Right Tool for the Right Market Environment

Characteristic	Mean Reversion	Trend Following
Win Rate	80-85%	40-50%
Average Trade Size	Small	Large
Trade Frequency	High	Low
Best Market	Range-bound, sideways	Trending, momentum
Worst Market	Strong trends	Choppy, range-bound
Holding Period	Days to weeks	Weeks to months

Key insight: Resows mean reversion tends to dominate on short-term timeframes (< 2 weeks), while trend following is more effective over longer-term periods (months+).

Prudent Risk Management is Non-Negotiable



1. Position Sizing

Strictly limit risk on any single trade to 1-2% of total capital. Position size should be determined by stop-loss distance, not conviction.



2. Stop-Losses

Use dynamic, volatility-adjusted stops (e.g., based on ATR) or statistical stops (e.g., 2.5 standard deviations from the mean). Time-based stops are also critical.



3. Signal Filtering

Avoid taking mean reversion trades during strong, established trends. Use a long-term moving average (e.g., 200-day SMA) as a trend filter.

Rigorous Validation Before Risking Capital: The Backtesting Gauntlet

Critical Requirements

- ✓ Test across multiple market regimes (low/high volatility, trending, range-bound).
- ✓ Ensure a minimum sample size of 100+ trades over multiple years.
- ✓ Use out-of-sample data to verify performance on unseen data.
- ✓ Measure key metrics: Sharpe Ratio, Max Drawdown, and Profit Factor.

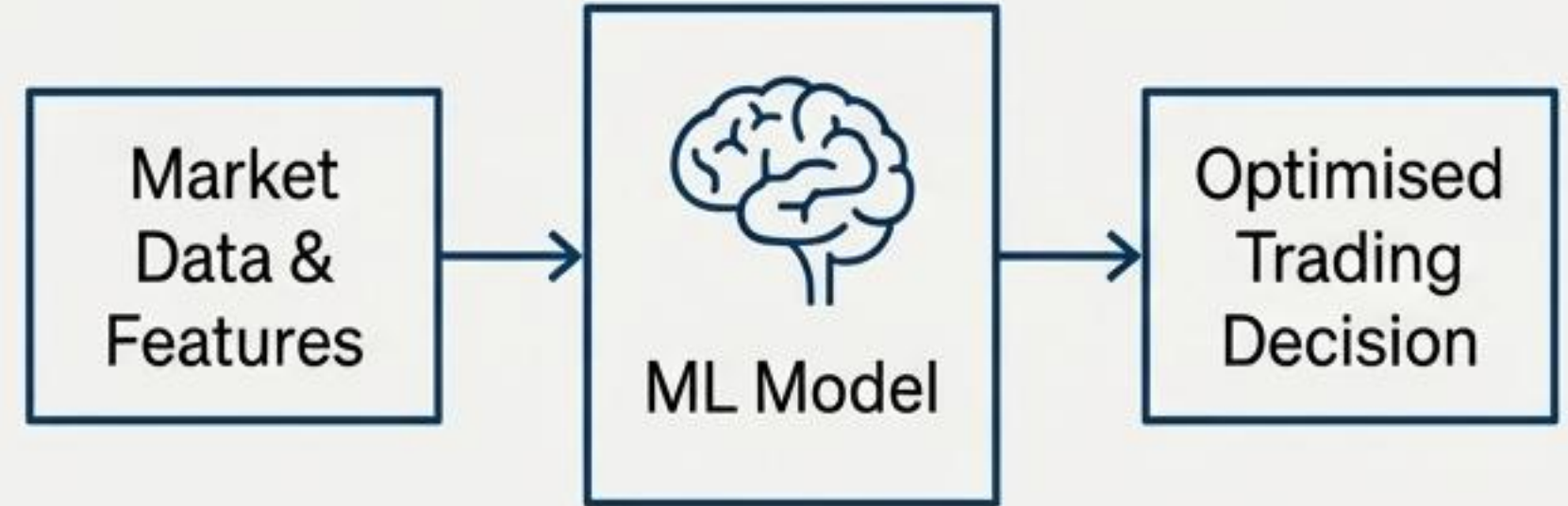
Common Pitfalls to Avoid

- ⚠ **Overfitting:** Creating rules that are too specific to historical data.
- ⚠ **Look-Ahead Bias:** Using future information in signal generation.
- ⚠ **Ignoring Costs:** Failing to account for commissions, slippage, and spreads.
- ⚠ **Assuming Constant Stationarity:** Markets evolve; re-test for mean reversion periodically.

The Frontier: Enhancing Strategies with Machine Learning

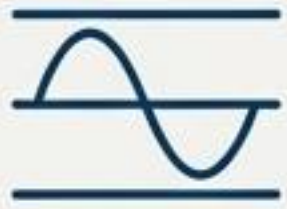
Two Key Approaches:

- **Supervised Learning:** Train models (e.g., Random Forest, Gradient Boosting) to predict the probability of mean reversion based on a wide range of features, allowing for probability-weighted position sizing.
- **Reinforcement Learning:** An agent learns optimal trading actions (buy, sell, hold) through trial and error, rewarding it for maximising portfolio returns while managing risk.



Powerful Example: A reinforcement learning agent can learn that a market is shifting from mean-reverting to trending, and adapt by holding positions longer to capture trend-based gains—a task at which static models fail.

Mastering Mean Reversion: Key Principles for Quantitative Success



1. Mean reversion profits from the predictable oscillation of prices around a central equilibrium, succeeding in range-bound markets where trends fail.



2. Success is built on a foundation of statistical rigour. Stationarity is a prerequisite, not a preference, validated by tools like the ADF test.



3. The toolkit ranges from direct Z-Score applications to sophisticated, market-neutral Pairs Trading strategies that isolate relative value.



4. Discipline separates amateurs from professionals. Robust risk management, rigorous backtesting, and adapting to change are the pillars of long-term success.